

Environmental taxes and the path to net-zero in the European Union

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Introduction

1. Description of our research question

Our research question is the following: “To what extent do environmental taxes reduce Greenhouse Gas Emissions (GHG) in EU countries, and what are their long-term socioeconomic impacts across economy?”. As the European Union commits to achieving net-zero emissions by 2050 under the European Green Deal, we aim to better understand the effectiveness and consequences of environmental taxation, both on emissions and socially.

To answer this question, we will use panel data for all EU member states. Our data set includes GHG emissions, GDP, environmental tax revenues, and several indicators of social well-being. These include the Gender Inequality Index (GII), a coefficient of human inequality, and a measure of energy poverty that is the inability to adequately heat one’s home. Together, these variables allow us to evaluate the distributive consequences of taxes.

After cleaning, harmonizing, and describing the data sets, we aim to estimate several relationships. First, we will measure the link between environmental taxation and GHG emissions. Do higher taxes lead to statistically significant reductions in emissions at EU level?

We will then investigate the association between environmental taxes and social inequality indicators. Our core question here can be summarized as: “Is there a social cost associated with environmental taxation, and if so, how large is it?” Specifically, we will analyse whether environmental taxes correlate with increases or decreases in gender inequality, income inequality, and energy poverty. This step is essential because policies that are environmentally effective but socially regressive may have a reduced long-term viability.

Building on these empirical relationships, we will construct forecasts for the period 2026–2050 to estimate the level of taxation required to meet the European Green Deal’s emissions targets. We consider a policy scenario relying solely on environmental taxation applied to relevant sectors. Our analysis will assess whether such an approach is socially and economically sustainable: How would different tax levels affect inequalities and energy poverty? How much revenue would such taxes generate, and would the required tax burden be compatible with each country’s GDP levels?

By combining environmental effectiveness, economic feasibility, and social impact, we can then address our overarching question: Is it socially and economically optimal for the EU to rely exclusively on environmental taxation to achieve net-zero emissions by 2050? If taxation alone proves insufficient or too socially costly, our results may highlight the need for complementary policies such as subsidies for green technologies or targeted social transfers to ensure a fair and effective ecological transition.

2. Link to Github

Our project is available on [Github](#).

3. Our data:

a. Environmental tax revenues

This data set reports annual environmentally related tax revenues for EU countries from the ([OECD](#)), with values in millions of USD covering from 1994 to 2023.

Name	Envt_Tax_Revenues
Number of rows	4050
Number of columns	36

b. GDP

This data set provides annual GDP figures for EU countries from the ([OECD](#)), measured in millions of USD, using exchange rate and covering from 1990 to 2024.

Name	GDP_and_AV
Number of rows	11100
Number of columns	44

c. Greenhouse gas emissions

This data set reports total greenhouse gas emissions for the 27 EU countries, in thousand tonnes of CO₂-equivalent. The data set is from ([Eurostat](#)) and covers from 1990 to 2023.

Name	GHG_Emissions
Number of rows	5508
Number of columns	11

d. Inability to keep home adequately warm

This data set collects the percentage of households reporting they are unable to keep their home adequately warm. We chose 3 households type: total, single female and single male; and 2 income situations: total and below 60% of the median (in situation of poverty). The data set is from ([Eurostat](#)) and covers 2003-2024.

Name	Warm_Households
Number of rows	5520
Number of columns	11

e. Human indexes

We focus on the Coefficient of Human Inequality and Gender Inequality Index (GII), from this ([UNDP HDR](#)) data set providing annual values for lots of Indexes, from 1990 to 2023.

Name	Soc_Indexes
Number of rows	206
Number of columns	1112

I. Modifying the data sets ¹

The data sets being quite large, we first remove variables we will not use and add aggregates that we think are important, before characterizing the variables we kept. Finally, to simplify our analysis, we merge all our sets in one data base.

1. Country keys

We construct a key that filtering the EU-27 countries and harmonizing the different identifiers, to ensure consistency.

2. Removing unwanted variables

For each dataset, we remove units, metadata, etc and systematically retain four essential components:

- Geographical identifier (ISO2, ISO3, or name).
- Reference year.
- Observed value (e.g., revenues, emissions, ...).

For the HDR index, we create two independent data sets, one for the Coefficient of Human inequality and another for the GII.

3. Adding aggregates

First, for the data sets that describe inequality indexes, we compute the European average across countries.

Secondly, for other economic data sets (GDP, Taxes, GHG emissions), we compute the European total across countries.

4. Modifying the characteristics of the variables

For the inability to keep housing adequately warm variable, we consider the household type variable and the income group variable as *factor*.

We rename variables to be able to merge our data sets; from “OBS_VALUE” appearing in several data sets to “name-of-dataset_value”.

We also rename the country identifiers as the ones in the country key we created.

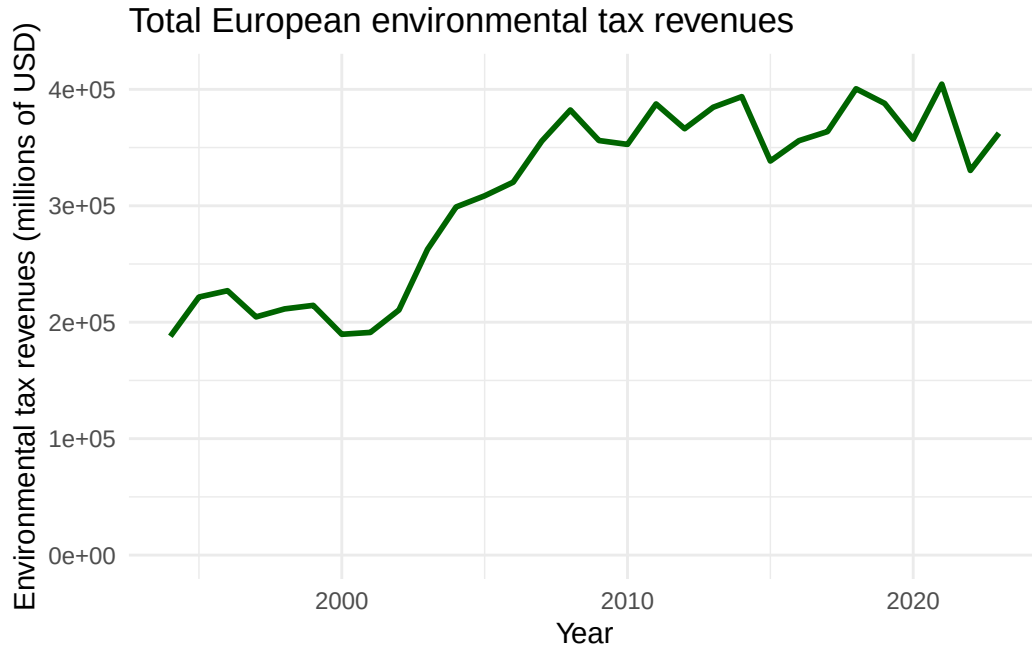
5. Merging

We can now merge all datasets into a single analytical framework on a country-year pair basis : we match on iso3 code, reference year, with *left_join()*.

¹For extensive description, please refer to the “*dm-project phase 2.qmd*” document.

II. Descriptive analyses ²

1. European environmental tax revenues



Descriptive statistics of the total European environmental tax revenues (all years combined)

Min	Max	Mean	Median	SD
187907.5	404466.1	310941.3	345576.7	76076.07

The total tax revenues increased from 190 billion USD in the late 1990s to 350 billion USD by 2010. Since then, the trend seems broadly stable, with noticeable fluctuations from year to year.

²For extensive analyses, please refer to the “*dm-project phase 2.qmd*” document.

2. European GDP

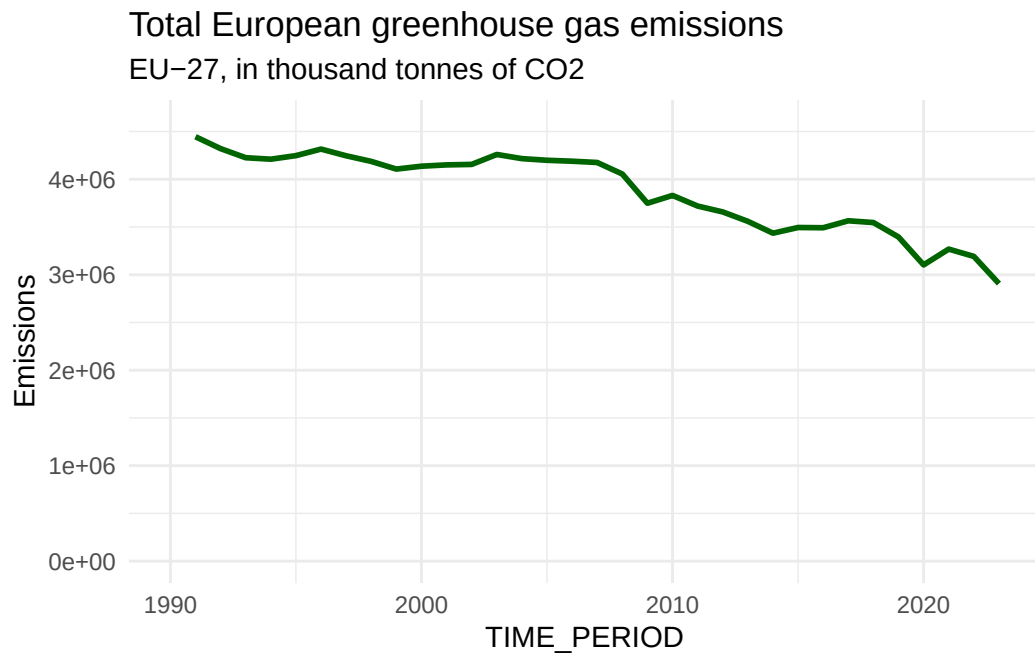


Descriptive statistics of the European GDP (all years combined)

Min	Max	Mean	Median	SD
6325442	19434510	12228913	13655502	4125193

European GDP has increased significantly from 6.3 trillion USD in 1990 to 19.4 trillion USD in 2024.

3. Greenhouse gas emissions

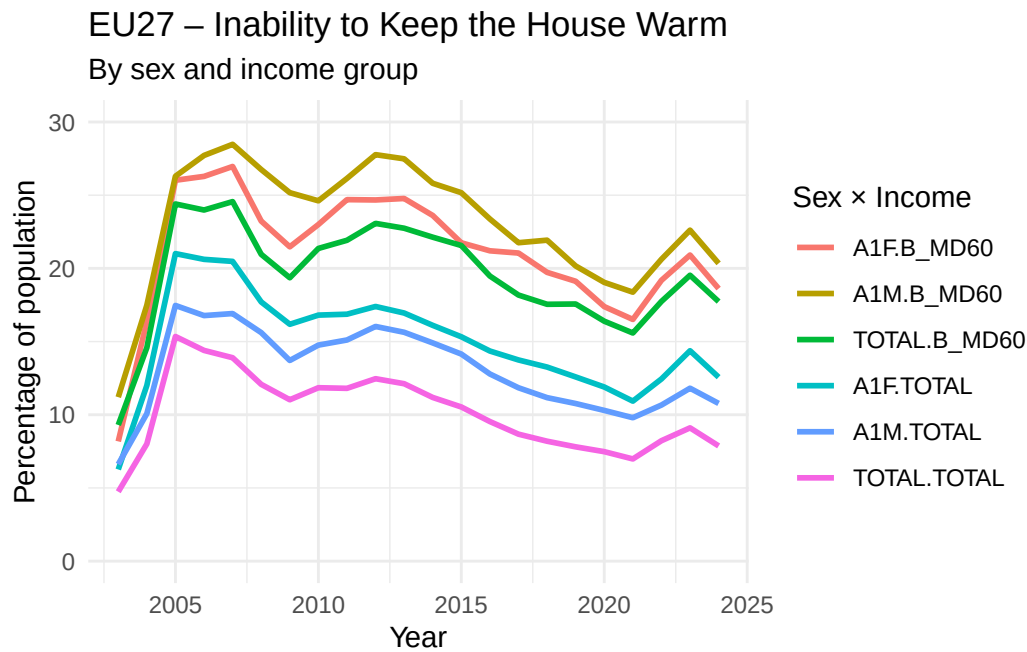


Descriptive statistics of EU total greenhouse gases emissions (all years combined)

Min	Max	Mean	Median	SD
2907204	4635248	3894053	4121922	434706.7

The data shows a clear a downward trend for the GHG: their maximum in 1990 is around 4.6 billion tonnes of CO₂-equivalent, and their minimum one in 2023 is around 2.9 billion tonnes of CO₂-equivalent.

4. Inability to keep home adequately warm per household type



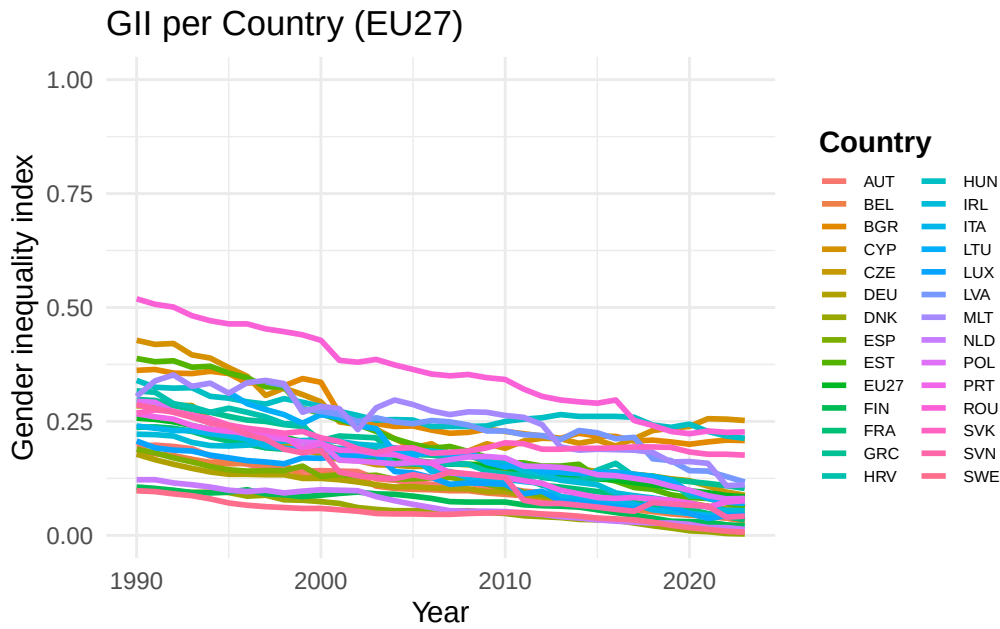
Descriptive statistics of the inability to keep house adequately warm, by household type (all years combined)

hhtyp	incgrp	Min	Max	Mean	Median	SD
A1F	B_MD60	8.18	26.96	21.12	21.34	4.25
A1F	TOTAL	6.26	21.01	14.99	14.85	3.51
A1M	B_MD60	11.20	28.48	23.10	23.99	4.25
A1M	TOTAL	6.62	17.47	13.08	13.25	2.88
TOTAL	B_MD60	9.30	24.56	19.53	19.50	3.68
TOTAL	TOTAL	4.74	15.35	10.15	10.03	2.69

The six curves, defined by sex and income, follow a similar pattern: a sharp rise around 2003, a gradual decline, and notable increases around 2009 and 2021.

Overall, the total population shows the lowest percentages, while households below 60% of the national median income show the highest. Among low-income single households, women report slightly higher rates than men (about 23.1% versus 21.1%).

5. GII per country³



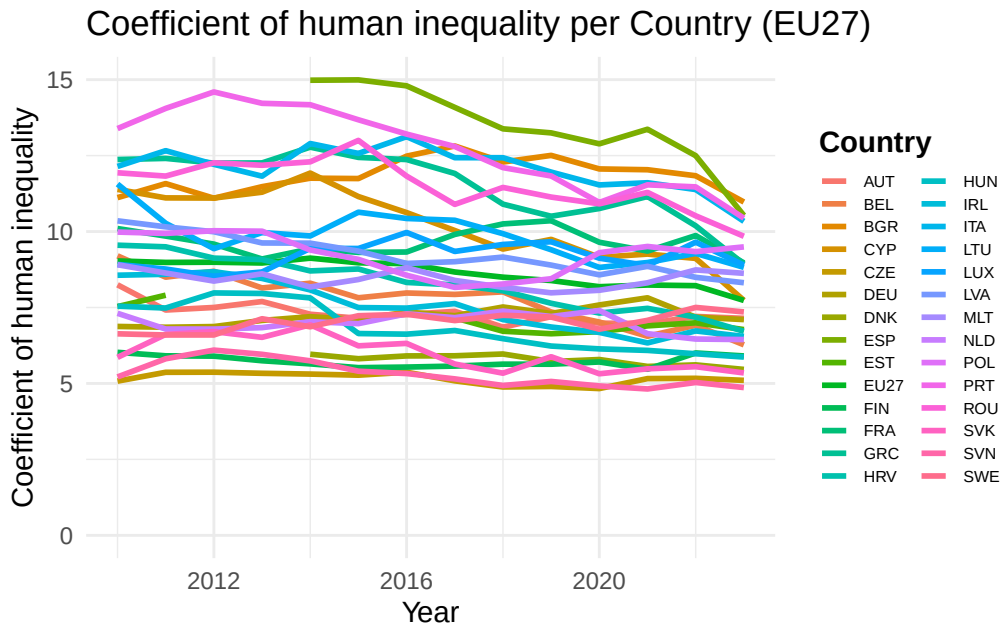
Descriptive statistics of the GII average

Min	Max	Mean	Median	SD
0.08	0.26	0.17	0.16	0.05

The Gender Inequality Index (GII) shows a clear downward trend across all European countries between 1990 and 2024. The EU27 average decreased from 25,7% to 0,08%, with a mean of 16,6% overall.

³For a more interactive and readable view of the graph in this section, please refer to the “*dm-project phase 3.html*” document.

6. Coefficient of human inequality per country ⁴



Descriptive statistics of the coefficient of human inequality average

Min	Max	Mean	Median	SD
7.73	9.12	8.64	8.81	0.43

The coefficient of human inequality varies widely across Europe. The EU27 average sits at $\approx 8.64\%$.

⁴For a more interactive and readable view of the graph in this section, please refer to the “*dm-project phase 3.html*” document.

III. Linear Regressions

1. Between taxes and GHG emissions

We answer one of our fundamental question: “Do taxes reduce GHG emissions ?”.

First we evaluate the coefficient between an increase in environmental taxes in the year n and the impact on GHG emissions in the same year n .

Regression Results

Dependent variable:	
GHG_Emissions_value	
Envt_Tax_Revenues_value	-1.029*** (0.254)
GDP_value	0.235*** (0.006)
TIME_PERIOD	-4,483.846*** (206.171)
Constant	9,044,371.000*** (414,867.900)
Observations	3,309
R2	0.846
Adjusted R2	0.846
Residual Std. Error	80,260.060 (df = 3305)
F Statistic	6,038.285*** (df = 3; 3305)
Note: *p<0.1; **p<0.05; ***p<0.01	

We estimated a linear regression to examine the relationship between environmental tax revenues and greenhouse gas emissions, while controlling for GDP and year. The model fits very well (Adjusted $R^2 = 0.8456$). Environmental tax revenues have a negative coefficient (≈ -1.03), indicating that higher environmental taxes are associated with lower emissions. The effect is statistically significant at the 1% level ($p < 0.001$). GDP enters positively, consistently with scale effects: larger economies produce more, and thus emit more; time has a negative coefficient, both highly significant. We do not calculate a log - level regression, because our following analysis and tax proposition will be made accordingly to levels of GHG.

Secondly, we try to evaluate the coefficient between an increase in taxes in the year n and the impact on ghg emissions in the following years $n+1$, to $n+5$.

Regression Results for GHG Emissions Leads 1-5 $p < 0.1$; $p < 0.05$; $p < 0.01$

Lead	term	Estimate	Std. Error	t value	p value	Signif
Lead 1	(Intercept)	8.884e+06	4.360e+05	2.038e+01	8.041e-87	***
Lead 1	Envt_Tax_Revenues	-1.516e+00	2.614e-01	-5.800e+00	7.280e-09	***
Lead 1	GDP	2.472e-01	6.064e-03	4.077e+01	2.414e-292	***
Lead 1	TIME_PERIOD	-4.405e+03	2.167e+02	-2.033e+01	2.033e-86	***
Lead 2	(Intercept)	8.846e+06	4.633e+05	1.909e+01	8.730e-77	***
Lead 2	Envt_Tax_Revenues	-1.970e+00	2.706e-01	-7.278e+00	4.307e-13	***
Lead 2	GDP	2.570e-01	6.348e-03	4.049e+01	2.310e-286	***
Lead 2	TIME_PERIOD	-4.387e+03	2.304e+02	-1.904e+01	2.071e-76	***
Lead 3	(Intercept)	8.710e+06	4.905e+05	1.776e+01	5.558e-67	***
Lead 3	Envt_Tax_Revenues	-1.989e+00	2.795e-01	-7.117e+00	1.389e-12	***
Lead 3	GDP	2.582e-01	6.575e-03	3.927e+01	5.899e-270	***
Lead 3	TIME_PERIOD	-4.320e+03	2.440e+02	-1.771e+01	1.195e-66	***
Lead 4	(Intercept)	8.891e+06	5.238e+05	1.697e+01	1.816e-61	***
Lead 4	Envt_Tax_Revenues	-2.083e+00	2.919e-01	-7.136e+00	1.234e-12	***
Lead 4	GDP	2.593e-01	6.894e-03	3.762e+01	3.168e-249	***
Lead 4	TIME_PERIOD	-4.411e+03	2.606e+02	-1.693e+01	3.645e-61	***
Lead 5	(Intercept)	9.172e+06	5.686e+05	1.613e+01	8.538e-56	***
Lead 5	Envt_Tax_Revenues	-2.012e+00	3.140e-01	-6.408e+00	1.755e-10	***
Lead 5	GDP	2.572e-01	7.414e-03	3.468e+01	5.290e-216	***
Lead 5	TIME_PERIOD	-4.552e+03	2.829e+02	-1.609e+01	1.586e-55	***

These lead regressions use GDP and a time trend as controls and all show a very strong fit, with adjusted R^2 remaining high (around 0.84) across all horizons.

Across the five specifications, the coefficient on environmental tax revenues is negative, statistically significant, and stable. It goes from -1.516 at horizon $n+1$ to approximately -2.0 at $n+2$ to $n+5$. This pattern suggests that higher environmental tax revenues are systematically associated with lower future GHG emissions.

All other coefficients are close to the ones seen before (GDP around 0.25, and time has coefficients -4.5 , all highly significant).

2. Between taxes and social inequalities

We estimate three regressions to evaluate the association between environmental tax revenues and three social outcomes, in percentage: energy poverty, human inequality, and gender inequality. In all models, we also add GDP and time (years). We chose to not include the Gender Inequality index and the Income inequality coefficient in one each others regression because of collinearity problems. Also, the coefficient of energy poverty is rather a consequence of both than a cause, so we do not include it in the other two as well.

The regression output reports some coefficients as **0.000**, but this reflects rounding given the precision of the results; the estimated value is in fact non-null and simply very small. The coefficients are provided in the following comments.

Regression Results

Dependent variable:	
Warm_Households_value	
Envt_Tax_Revenues_value	0.00004** (0.00002)
GII_value	51.246*** (3.455)
coef_ineq_value	3.104*** (0.090)
hhtypA1M	0.124 (0.484)
hhtypTOTAL	-3.309*** (0.484)
incgrpTOTAL	-9.123*** (0.395)
log(GDP_value)	-2.083*** (0.242)
TIME_PERIOD	0.041 (0.052)
Constant	-66.602 (105.679)
Observations	2,100
R2	0.600
Adjusted R2	0.598
Residual Std. Error	9.055 (df = 2091)
F Statistic	391.866*** (df = 8; 2091)
Note: *p<0.1; **p<0.05; ***p<0.01	

In the model explaining energy poverty, we include the income-inequality coefficient. The resulting model has strong explanatory power (Adjusted $R^2 \approx 0.60$). The coefficient on environmental tax revenues remains small, positive, stable (+3.7e-05) and significant. Also, the income-inequality coefficient shows strong positive and significant associations with energy poverty.

Regression Results

Dependent variable:	
coef_ineq_value	
Envt_Tax_Revenues_value	0.00003*** (0.00000)
log(GDP_value)	-0.255*** (0.057)
TIME_PERIOD	-0.081*** (0.013)
Constant	175.186*** (25.766)
Observations	2,114
R2	0.052
Adjusted R2	0.051
Residual Std. Error	2.324 (df = 2110)
F Statistic	38.750*** (df = 3; 2110)
Note: *p<0.1; **p<0.05; ***p<0.01	

For human inequality, the model remains statistically valid but with limited explanatory power (Adjusted $R^2 \approx 0.05$), and environmental taxes have a small impact on inequality (+3.0e-05) but is highly significant. GDP and time both show negative effects.

Regression Results

Dependent variable:	
GII_value	
Envt_Tax_Revenues_value	0.00000*** (0.00000)
log(GDP_value)	-0.031*** (0.001)
TIME_PERIOD	-0.004*** (0.0002)
Constant	7.639*** (0.308)
Observations	3,307
R2	0.405
Adjusted R2	0.405
Residual Std. Error	0.059 (df = 3303)
F Statistic	750.631*** (df = 3; 3303)
Note: *p<0.1; **p<0.05; ***p<0.01	

Similarly, in the gender-inequality model the environmental-tax coefficient remains positive (+6.9e-07) and highly significant. GDP and time effects are negative and the model is substantially strong (Adjusted $R^2 \approx 0.40$).

3. Between tax and economy

Regression Results

Dependent variable:	
log(GDP_value)	
Envt_Tax_Revenues_value	0.0001*** (0.00000)
TIME_PERIOD	0.025*** (0.002)
Constant	-38.716*** (4.875)
Observations	3,309
R2	0.630
Adjusted R2	0.630
Residual Std. Error	0.950 (df = 3306)
F Statistic	2,814.265*** (df = 2; 3306)
Note: *p<0.1; **p<0.05; ***p<0.01	

The model shows that environmental tax revenues are positively associated with GDP, and the relationship is highly statistically significant, although the effect size is very small (≈ 0.0001 per unit of tax revenue). The time trend is positive and highly significant, reflecting steady GDP growth over the period.

It also has substantial explanatory power (Adjusted $R^2 \approx 0.63$). However, the relationship of GDP with taxes is correlational and does not imply causation, it indicates that environmental tax revenues tend to increase alongside GDP, consistent with the idea that wealthier economies can collect more from climate-related taxes.

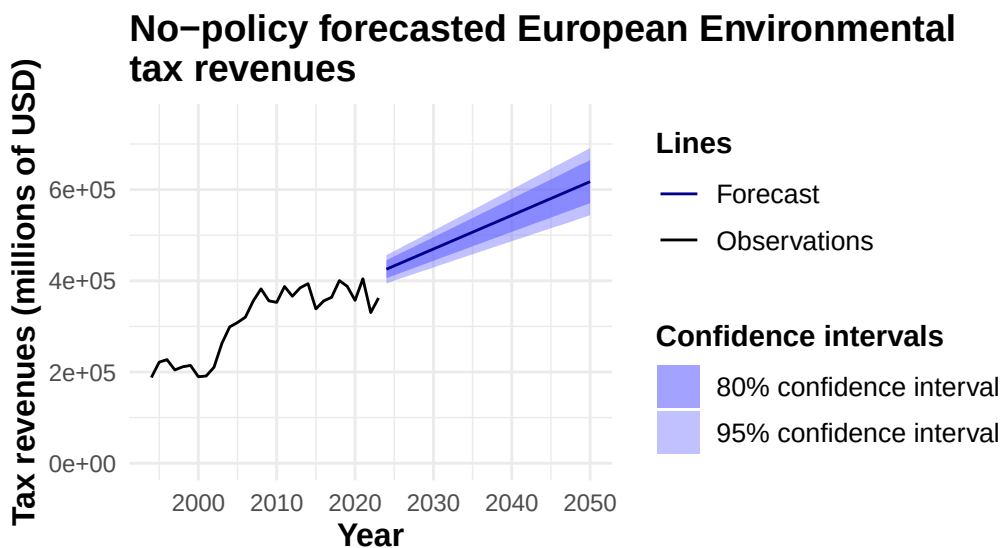
IV. Forecasting: no-policy

We forecast the impact of environmental taxes on various variables, in a scenario where no policy is implemented.

1. Forecasted environmental tax revenues

Regression Results

Dependent variable:	
Envt_Tax_Revenues_value	
TIME_PERIOD	7,384.111*** (848.385)
Constant	-14,520,045.000*** (1,703,998.000)
Observations	30
R2	0.730
Adjusted R2	0.720
Residual Std. Error	40,220.080 (df = 28)
F Statistic	75.755*** (df = 1; 28)
Note: *p<0.1; **p<0.05; ***p<0.01	



If no new tax policy is implemented, environmental tax revenues in the EU27 are projected to increase from 362.4 billion USD in 2023 to 617.4 billion USD in 2050, representing a 70% growth based on an extrapolation of current trends. This steady upward trajectory suggests continued expansion of green taxation

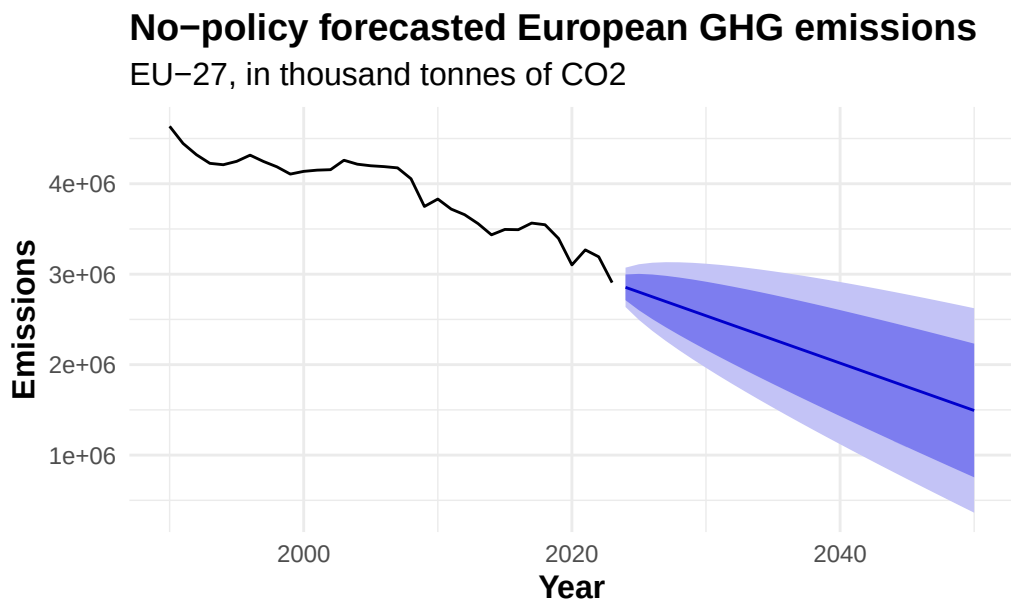
policies, however, it's important to note that this forecast assumes the continuation of existing trends under a no-change-in-policy scenario.

2. Greenhouse Gas emissions

	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
2024	2854839	2712621.1	2997057	2637335.4	3072343
2050	1493350	754363.6	2232337	363167.7	2623533

ARIMA(0,1,0) with drift: Summary

Term	Estimate	Std..Error
drift	-5.236e+04	1.902e+04
sigma^2	1.232e+10	
log likelihood	-429.68	
AIC	863.36	
AICc	863.76	
BIC	866.35	



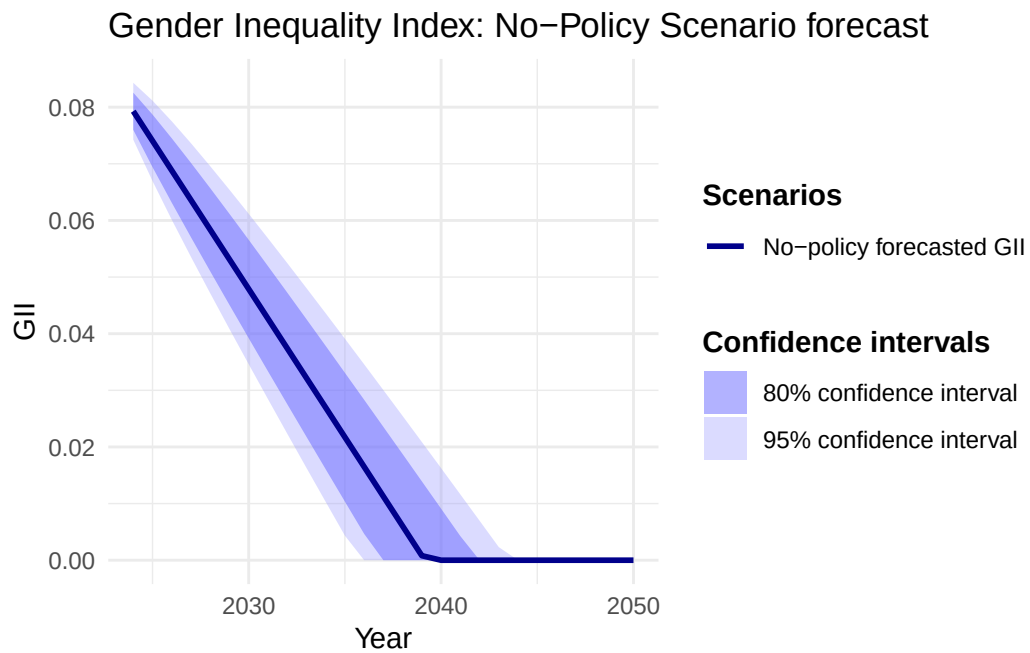
This ARIMA-based forecast suggests a slow but steady decline in total EU27 greenhouse gas emissions from 2024 to 2050. The ARIMA(0,1,0) with drift implies an average annual reduction of about 52 million tonnes of CO₂-equivalent, with emissions falling from roughly 2.85 billion tonnes in 2024 to around 1.49 billion tonnes in 2050.

Despite this downward trend, projected emissions in 2050 remain far above zero—even in the most optimistic scenario—indicating that current trends are insufficient to reach EU net-zero targets (about 363 million tonnes of CO₂-equivalent).

3. Social inequalities:

a. Gender Inequality Index

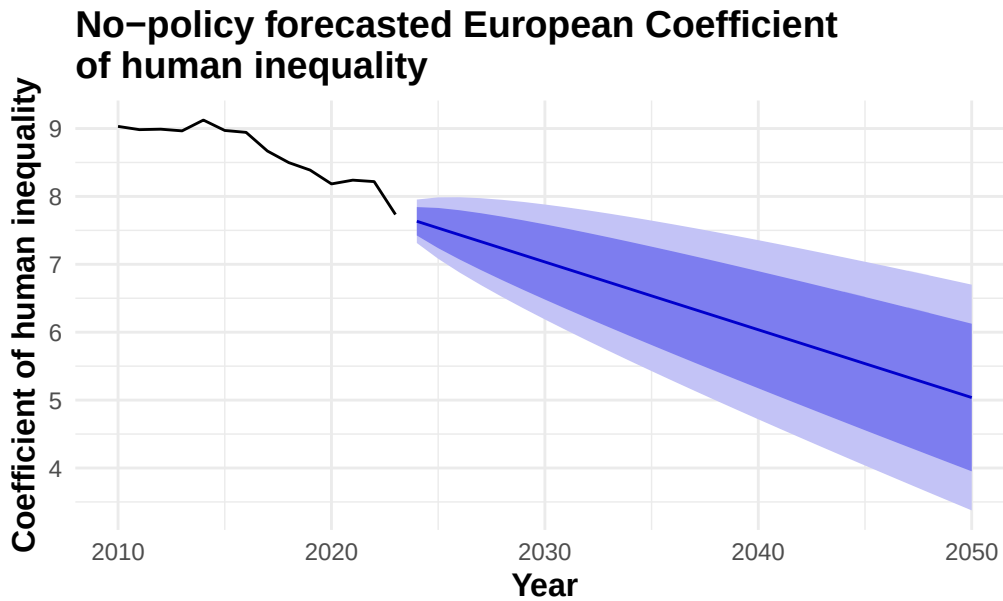
	year	mean	lower80	upper80	lower95	upper95
1	2024	0.07928575	0.07600342	0.08256807	0.07426586	0.08430563
2	2050	0.00000000	0.00000000	0.00000000	0.00000000	0.00000000



The GII is forecast to decline from 0.079 in 2024 to zero by 2050 (with negative values cut, as the index ranges from 0 to 1).

b. Coefficient of Human Inequality

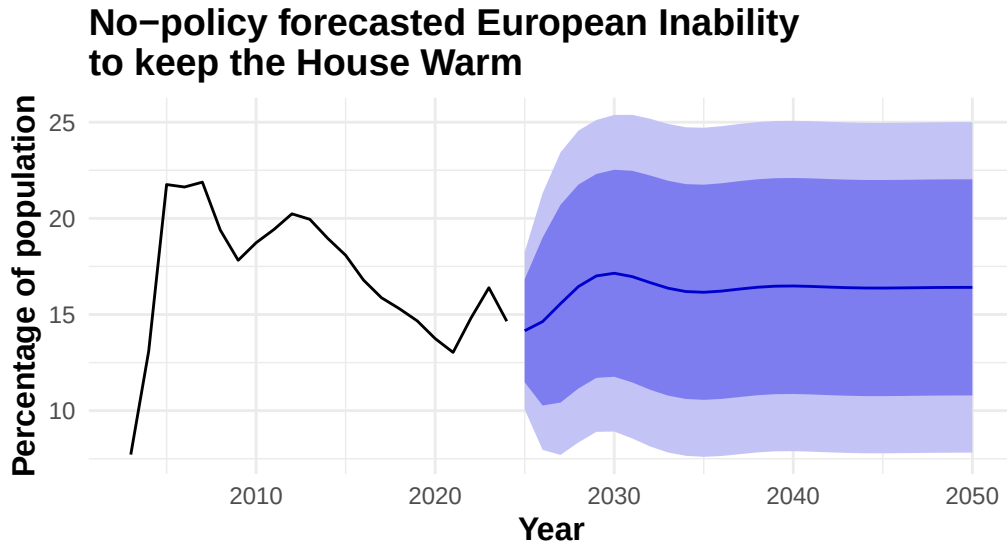
	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
2024	7.633930	7.424743	7.843117	7.314006	7.953854
2050	5.038047	3.951079	6.125015	3.375673	6.700421



Human inequality is expected to fall from 7.63 to about 5.04 over the same period, suggesting gradual improvement, though uncertainty increases over time.

c. Energy poverty

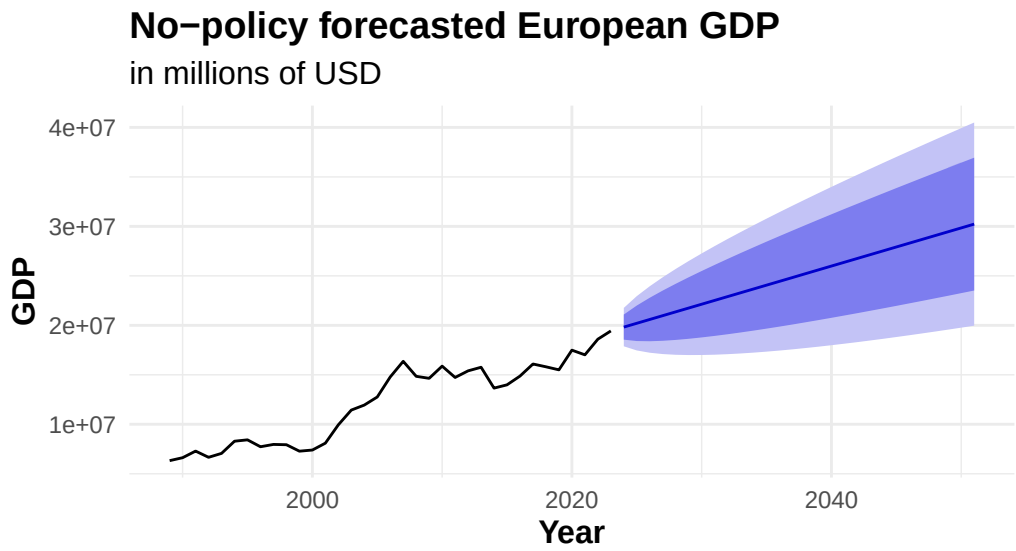
	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
2025	14.15495	11.47802	16.83189	10.060934	18.24897
2050	16.41007	10.79167	22.02848	7.817468	25.00268



In contrast, the share of households unable to heat their homes is projected to remain broadly stable, edging up slightly from 14.15% in 2024 to 16.41% in 2050, with wide confidence intervals.

3. Gross Domestic Product

	Point Forecast	Lo 80	Hi 80	Lo 95	Hi 95
2024	19820071	18551674	21088468	17880225	21759917
2051	30230213	23518486	36941939	19965513	40494913



The EU27 aggregate GDP is projected to grow steadily from 19.4 trillion USD in 2024 to 29.5 trillion USD in 2050, representing approximately 50% growth over the period, though the widening confidence intervals indicate increasing uncertainty in long-term projections.

V. Forecasting: net-zero emission target

We are trying to evaluate the needed level of tax to reach the Zero net emissions by 2050 desired by the European Union, as if there only was a tax policy available; then we check the impact of this level of tax on the social inequalities and on the economy.

1. Deriving a realistic tax path to reach net-zero by 2050

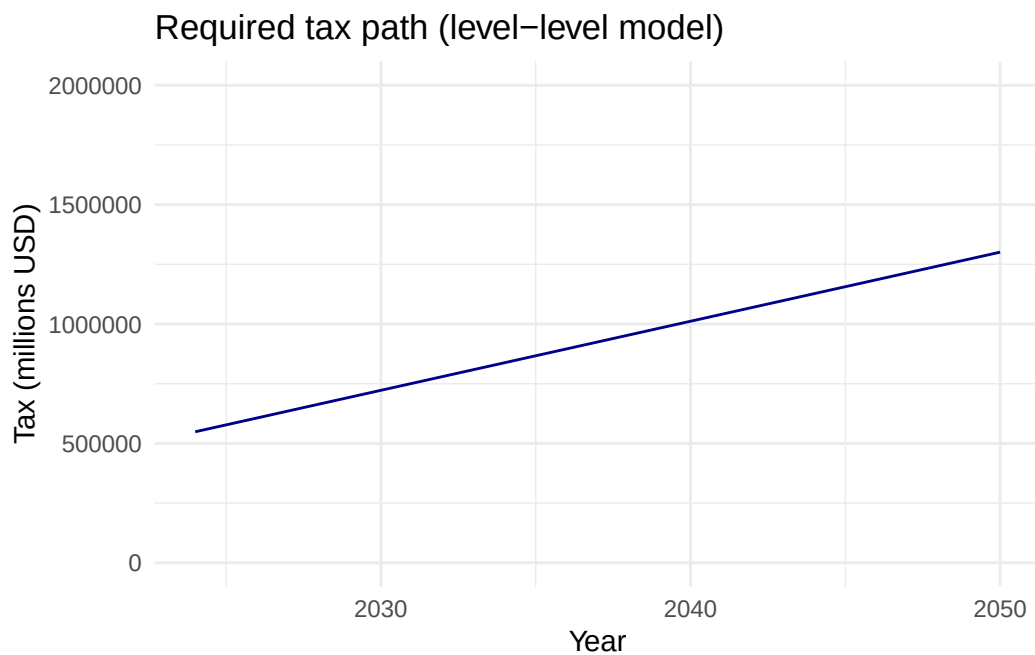
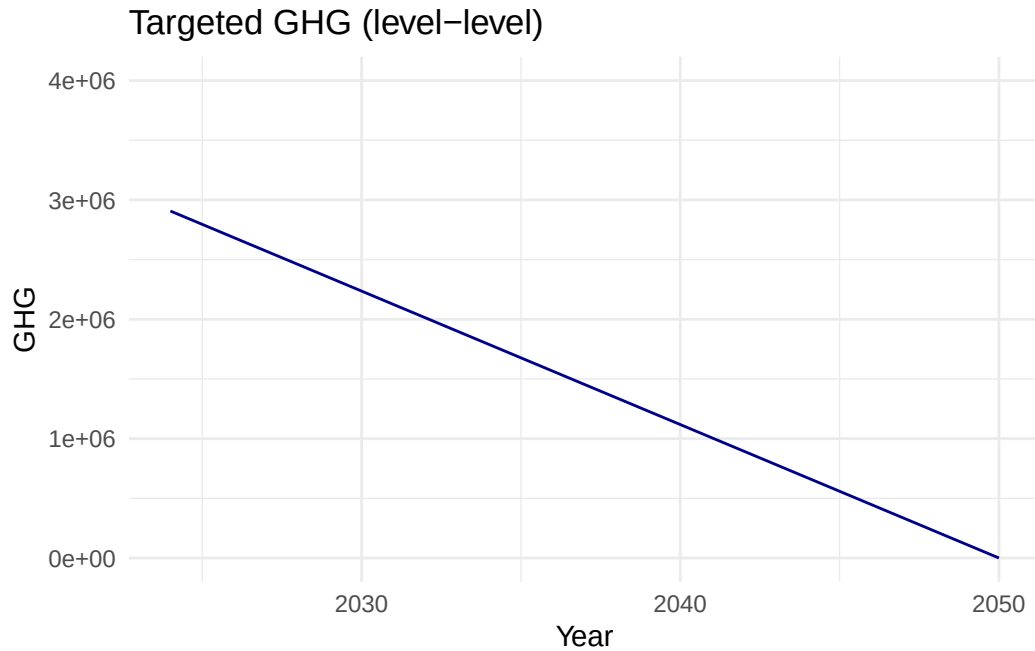
a. The forecast model

We use a linear regression relating aggregate GHG emissions to Environmental tax revenues at the EU-27 level. The 2023 tax level is anchored to the last observed value.

Regression Results

Dependent variable:	
GHG_total	
Envt_Tax_Revenues_value	-3.864*** (0.721)
Constant	5,027,191.000*** (230,626.400)
Observations	30
R2	0.506
Adjusted R2	0.489
Residual Std. Error	295,435.400 (df = 28)
F Statistic	28.712*** (df = 1; 28)
Note: *p<0.1; **p<0.05; ***p<0.01	

The estimated regression reveals a statistically significant negative relationship between environmental tax revenues and total GHG emissions (-3.86 , $p\text{-value} < 0.001$). This indicates that, historically, an increase of one million USD in environmental tax revenues is associated with a reduction of approximately 3.9 thousand tonnes of CO₂-equivalent in aggregate emissions. The model explains a meaningful share of the observed variation in emissions, with an Adjusted R^2 of 0.49.



Using this estimated relationship, we construct a counterfactual emissions trajectory from approximately 2.9 billion tonnes in 2023 to net-zero in 2050. After 2024, the tax path follows a linear trajectory toward 2050, reflecting the progressively stronger fiscal effort needed to reduce emissions to zero.

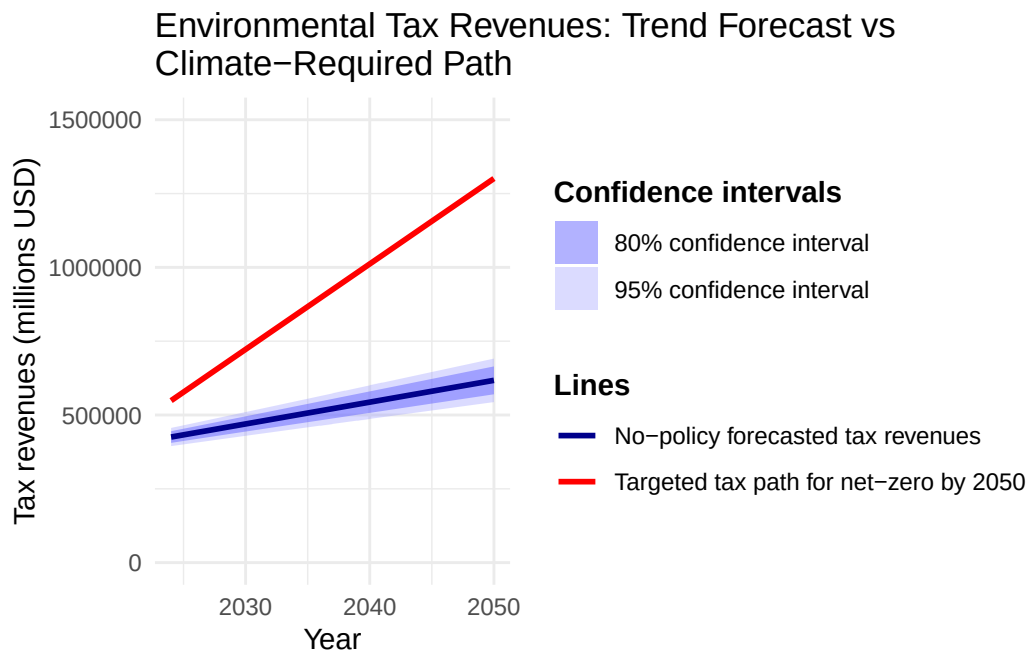
According to the tax path, environmental tax revenues would need to rise from 550 billion USD in 2024 to about 1.3 trillion USD in 2050, highlighting both the scale of the fiscal effort required and the limitations of relying solely on environmental taxation, given the modest explanatory power of the reduced-form regression.

b. Comparing with the no-policy forecast

We evaluate the gap between historical trends and climate targets before assessing the implications of the required tax paths.

i. Tax policy: targeted vs predicted

To assess the adequacy of the policy, we compare the forecasted path of total environmental tax revenues based on historical trends, with the one required to reach net-zero emissions by 2050.

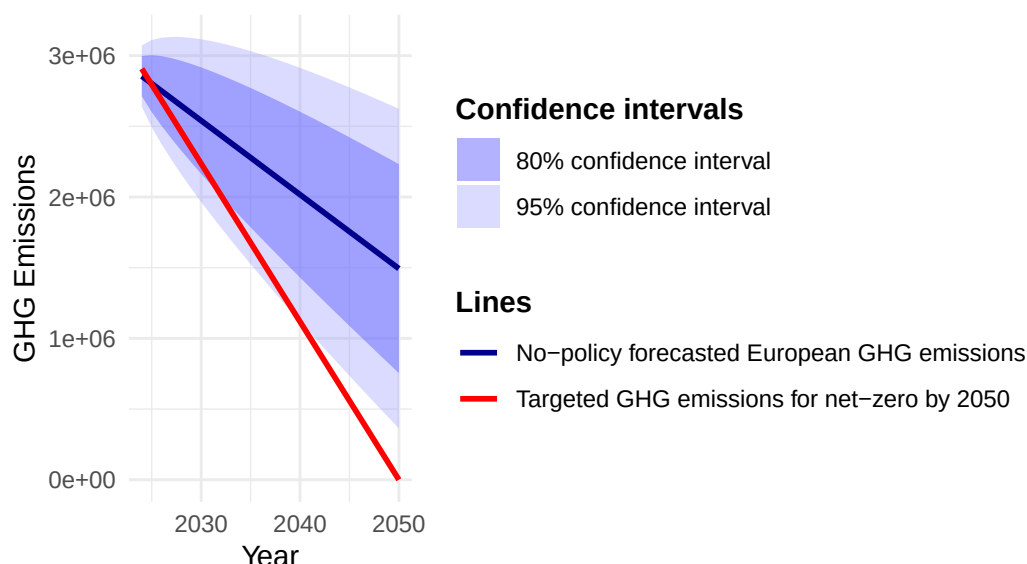


Comparing the two paths shows a significant and growing shortfall. The historical trend regression predicts that total environmental tax revenues would increase by approximately 7.3 billion USD per year and in contrast, the required one to achieve net-zero calls for an annual increase of roughly 28.9 billion USD per year. From the first year (2024), forecasted revenues fall well below the required level, and by 2050 the gap widens to roughly 684 billion USD. These results highlight that in order to achieve the net-zero emissions target, environmental taxation would need to be increased rapidly.

ii. GHG emissions: targeted vs predicted

We can compare the emissions trajectory implied by a continuation of historical tax trends (ARIMA seen before) with the counter-factual path required to achieve net-zero emissions by 2050.

Greenhouse gases emissions: Trend Forecast vs Climate-Required Path



As the graph shows, achieving the GHG emissions target would require significant, but not dramatic, changes to current tax practices. From around 2035 onwards, the required tax-path falls outside the 95% confidence interval, indicating that the probability of reaching the targeted emissions level is very low.

2. Is that policy fair? Its impact on social inequalities

a. The forecast model

We project the future evolution of our three inequality indicators: average warming inability, the coefficient of human inequality, and the Gender Inequality Index (GII).

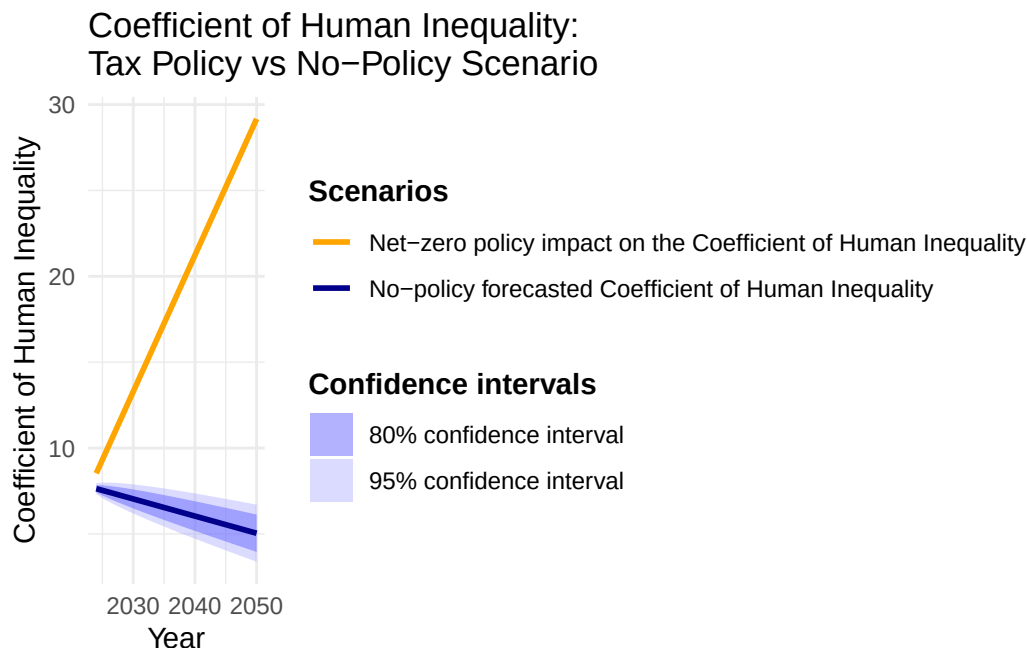
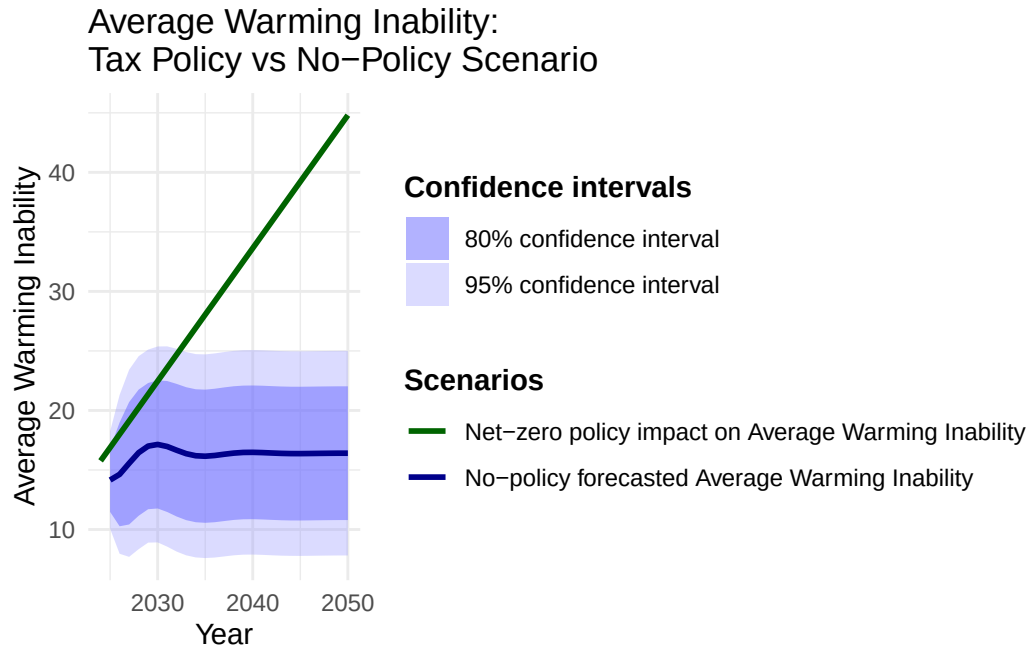
Starting from their last observed values, we simulate paths up to 2050 by combining the estimated marginal effects of environmental tax revenues and time trends obtained from the regressions. The required path is expressed as an average annual increase, applied uniformly over time. The projections are established by applying these marginal effects cumulatively year after year. For the GII, predicted values are constrained to remain within 0 and 1.

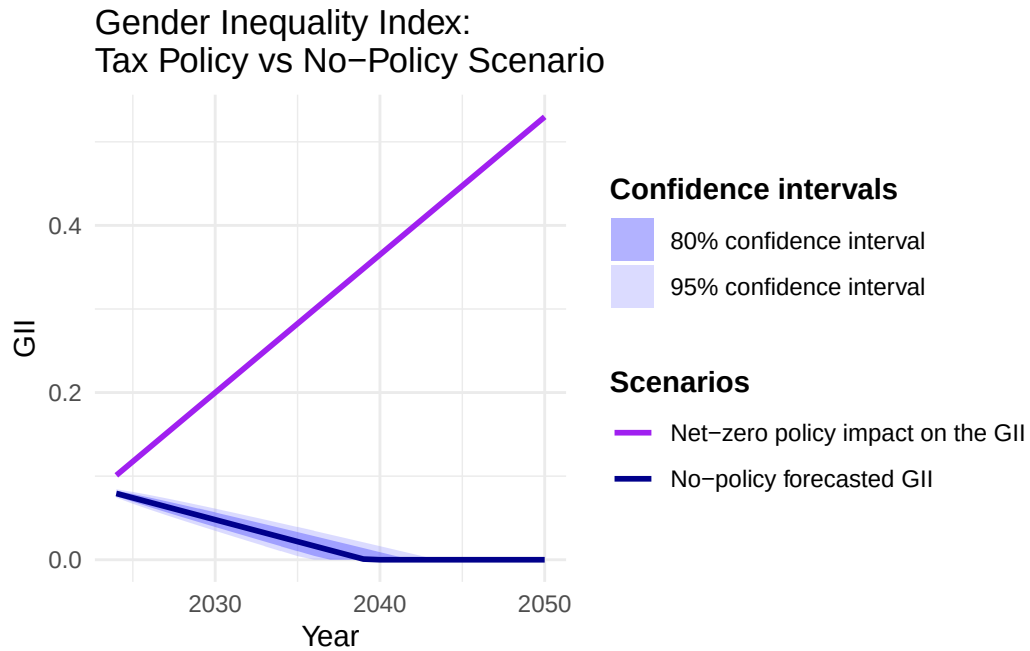
Year	Predicted Warming	Predicted Inequality	Predicted GII
2024	15.767	8.528	0.101
2050	44.803	29.166	0.530

Projections show that the required trajectory significantly increases the three inequality indicators over the period 2024-2050. Indeed, the average warming inability rises from 15.8% of the population in 2024 to 44.8% in 2050, the human inequality coefficient increases from around 8.5 to 29.2, indicating a considerable widening of social disparities. Similarly, the gender inequality index will increase from around 0.10 to around 0.53.

b. Comparing with the no-policy forecast

To assess how the proposed environmental tax policy affects social inequalities relative to a baseline scenario without policy intervention, we compare the projected evolution of each indicator under the tax trajectory with their ARIMA-based forecasts in the absence of environmental taxation.





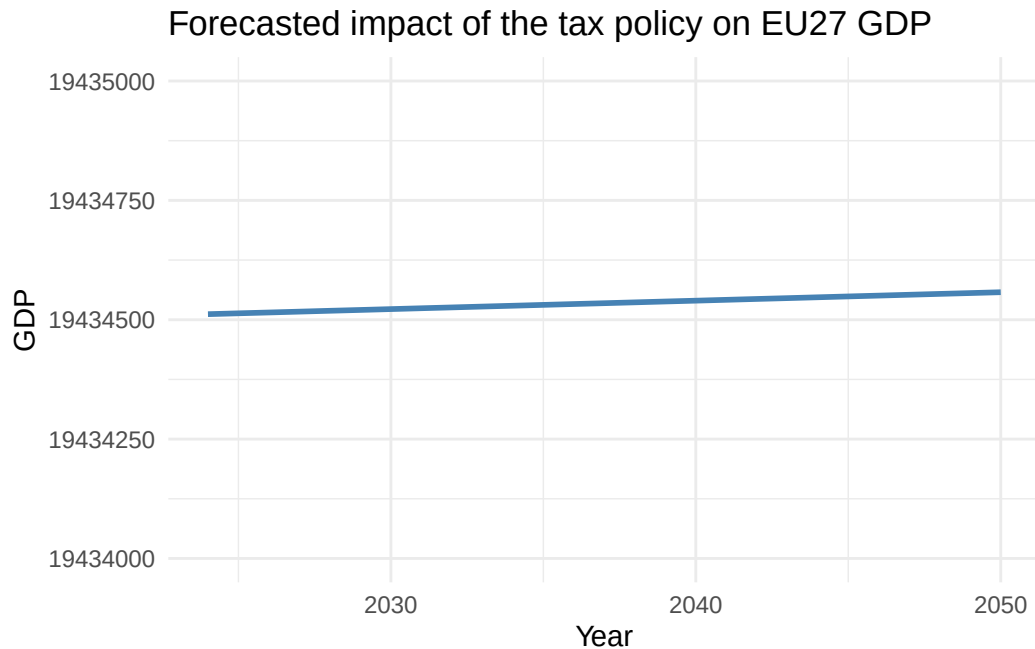
The comparison between the net-zero target and no-policy scenarios highlights a clear trade-off. By contrast of the sharp increases seen before, without the required tax policy, warming inability remains around 16–17%, human inequality gradually declines, and gender inequality decreases over time.

These projections suggest a potential trade-off between ambitious climate policy and social equity outcomes. Complementary redistributive or compensatory measures should be considered.

3. Is that policy viable ? Its impact on the GDP

To evaluate the potential economic impact of the proposed environmental tax policy, we forecast EU27 GDP under the tax scenario using the estimated coefficients from our GDP regression model. We construct the future path by applying the average annual increase in environmental tax revenues, combined with the underlying time trend, cumulatively on the last observed GDP level.

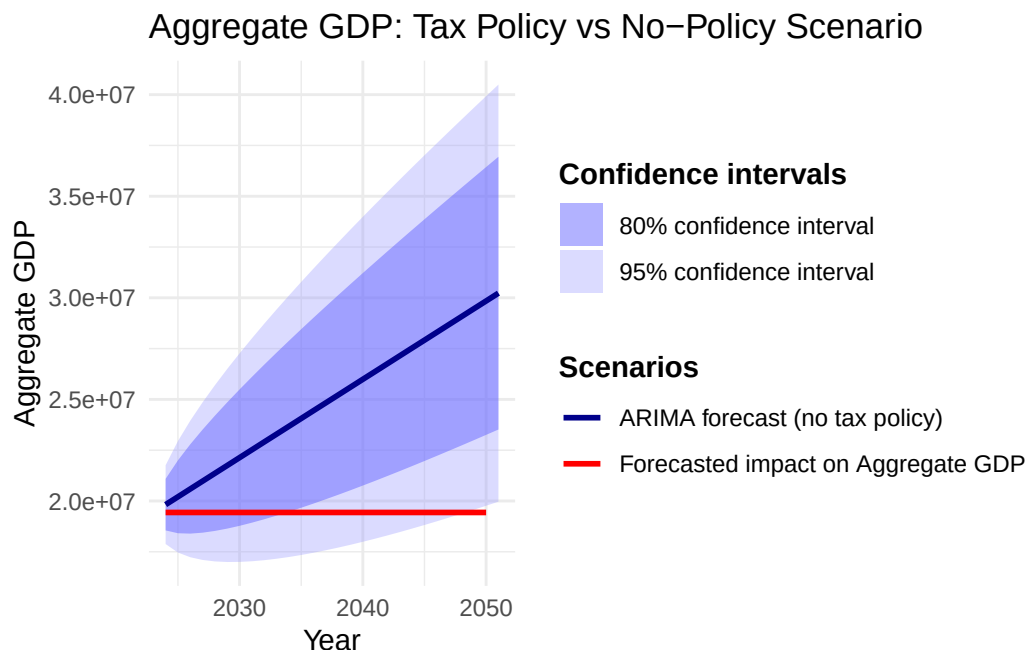
a. The forecast model



Under the tax policy scenario, projected GDP exhibits an almost flat trajectory over the period 2024–2050, increasing from approximately 19,434,512 million USD to 19,434,558 million USD. This corresponds to an average annual growth rate close to zero, indicating a near-complete stagnation of economic activity over the projection horizon. This result suggests that the required tax trajectory would substantially constrain aggregate economic growth. While these projections should not be interpreted as causal estimates, they point to a potentially significant macroeconomic cost associated to the the tax policy.

b. Comparing with the no-policy forecast

We then project this GDP path alongside the ARIMA-based GDP forecast in the absence of any tax policy, allowing us to visually and quantitatively compare how the tax might influence aggregate econ



The comparison shows that the GDP forecast without the tax policy is consistently higher than the GDP projected under the tax scenario, with the no-policy path rising to 29.5 trillion USD in 2050, while the tax policy scenario remains nearly flat around 19.4 trillion USD. This indicates that, according to our model, implementing the environmental tax would largely slow the growth of aggregate GDP compared with the baseline. However, the tax-induced impact is relatively small in absolute terms early on, suggesting limited short-term economic disruption.

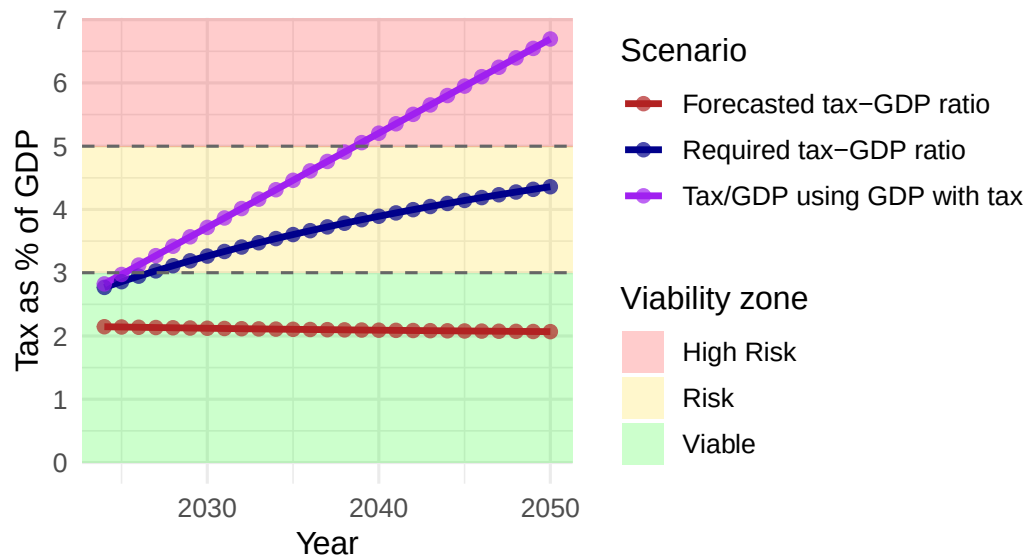
c. Ratio Tax/GDP

To further assess the economic viability of the environmental tax policy, we examine the evolution of the tax burden relative to aggregate economic output by computing the environmental tax as a share of GDP under three distinct scenarios.

First, we calculate the required-tax-to-GDP ratio, obtained by dividing the required tax trajectory to reach the net-zero objective by the baseline GDP forecast. Secondly, we compute a predicted-tax-to-GDP ratio, based on the forecasted environmental tax path in the absence of the policy, again expressed relative to forecasted GDP. Finally, we construct a tax-to-GDP ratio using GDP projected under the tax policy itself, which accounts for the potential macroeconomic feedback of the tax on economic activity.

Economic Viability of Environmental Tax Policy

Tax as % of GDP under three scenarios



These three trajectories are plotted against viability thresholds at 3% and 5% of GDP, defining zones of relative economic feasibility or risk.

The comparison of tax-to-GDP ratios across the three scenarios shows the forecasted-tax ratio remains in the viable zone, while the required and the ratio with the GDP-under-policy increase over time respectively in the Risk and High risk zones.

This framework allows us to assess how the interaction between taxation and economic output may affect the policy's sustainability, including risks of growth disruption, reduced political acceptability, or broader macroeconomic instability.

Conclusion

In conclusion, this research project has evaluated the effectiveness of environmental taxation as a tool for achieving the European Union’s goal of net-zero emissions by 2050. The study involved cleaning and merging large datasets covering the EU-27 from 1990 to 2024. By integrating indicators such as environmental tax revenues, the Gender Inequality Index (GII), the Coefficient of Human Inequality, and energy poverty, we assessed both environmental efficacy and socioeconomic sustainability.

Our findings indicate that while environmental taxes have a statistically significant negative association with GHG emissions, the impact has been small and we revealed that the current forecasted tax trajectory is insufficient to reach the European Green Deal’s objectives without important negative socioeconomic consequences.

To determine the tax level required for the 2050 target, we inverted a level-level linear regression of total GHG emissions against tax revenues. This model allowed us to find a “required tax path” that would theoretically drive emissions to zero by itself. We found that environmental tax revenues would need to scale up to roughly 1.3 trillion USD by 2050 (\$≈\$684 billion USD more than predicted historical trends).

The impacts of such a massive tax increase on society and the economy would be severe:

- **Social Inequality:** The regressions show that environmental taxes are positively correlated with energy poverty, income inequality, and gender inequality. Implementing the required tax path would likely exacerbate these social costs, and increase these indicators largely above the levels expected in a no-policy scenario.
- **Economic Viability:** Although tax revenues tend to increase alongside GDP, the required fiscal burden would likely reach above 3% of the total GDP. This exceeds empirical thresholds for economic stability and could be highly disruptive.

If relying exclusively on environmental taxation proves to be socially regressive or economically unviable, what alternative strategies could the European Union use to fill the gap? Some possibilities could be the use of complementary green technology subsidies, the implementation of stricter sector-specific regulations, or the potential for economic sobriety and degrowth to reduce the overall energy demand.

Limitations

This study has several limitations that should be considered when interpreting the results. First, the analysis focuses on environmental taxation as if it were the sole policy instrument, abstracting from other climate policies implemented at the EU level. Future research could address this by jointly modelling environmental taxes alongside complementary instruments such as subsidies, regulations, and redistributive measures within an integrated policy or general equilibrium framework.

Second, data availability limits the inequality analysis, as the Coefficient of Human Inequality is only available for the period 2010–2023. As longer time series become available, future studies could extend the estimation horizon or combine aggregate indicators with micro-level data to strengthen the robustness of inequality projections.

Third, although detailed information by household type and income group exists for energy poverty, this heterogeneity is not exploited, and the EU-level aggregation may mask substantial cross-country differences. Future work could rely on household-level microsimulation or country-specific analyses to better capture distributional effects.

Finally, sector-level impacts are not analysed due to inconsistencies in sector classifications, a limitation that could be overcome by using harmonized input–output data or computable general equilibrium models to study sectoral adjustment mechanisms.